Raspberry Pi Pico

P1
rpi pico 1
2648-SC0915CT-ND
tht

R10
220k

R1
220
3V1
green

D1
D_Schottky

R2
220
D13
blue

J1

J3
joystick

J4
speed knob

NT5 NetTie_2
NT6 NetTie_2
NT7 NetTie_2
NT8 NetTie_2

left motor

U1
DRV8245HQPWPRQ1

LVL6, in independent mode that means faults are retries

C1
0.1uF

R7
8.2k +-10% LVL2 INDEPENDENT

HSL1
Heatsink

J100
0197054001

J101
0197054001

optionally replace J7 to be weelye compatible

J7
1017491

U2
DRV8245HQPWPRQ1

C2
0.1uF

R6
8.2k +-10% LVL2 INDEPENDENT

HSR1
Heatsink

J8
1017491

optionally replace J8 to make the PCB weelye compatible

wellye style
J104

wellye RX25 style
J105

ADC (for measuring battery voltage and motor current)

U4
TLA2024IRUGR

R13
100k

R14
13.7k

R11
604

R12
604

TP0
TestPoint

TP1
TestPoint
left_ipropi

TP2
TestPoint
right_ipropi

C4
0.1uF

R20
EXB34V182JV
pullup resistors
1.8k

qwiic I2C plug

J2
Conn_01x04

weelye compatible header

J11
Conn_01x10_Pin

D2
D_Schottky

3.5mm audio jacks for if you want to connect buttons as an alternative to the joystick

B1
ASJ-200-B-HT

B2
ASJ-200-B-HT

B3
ASJ-200-B-HT

B4
ASJ-200-B-HT

battery power circuit

J102
0197054001

optionally replace J5 to be weelye compatible

J103
0197054001

J5
TB005-762-02BE

R3
8.2k

C5
33uf

SW1
screw terminal

R4
100k

18V zener
D3

PSMN1R2-25YLDX
Q1
reverse polarity protection

PSMN1R2-25YLDX
Q2
on/off switch

J6

alternative voltage regulator

Instead of the voltage regulator circuit, you could get a single THT module that outputs 5V.
Make sure the pinout matches.
Make sure the min and max input voltage are compatible with your battery.

U5
D78B05T, RBT05W24S05, pololu5593

voltage regulator circuit https://webench.ti.com/appinfo/webench/scripts/SDP.cgi?ID=F7F7D983B50C08E2

U3
TPS54202HDDCR

R5
100k

C3
10uF

C6
0.1uF

L1
12uH

C7
56pF

R8
100k

R9
13.7k

C8
47uF

3mm diameter bolt holes
3.35in x 2.2in
85.09mm x 55.88mm

H1 M3
H2 M3
H3 M3
H4 M3

https://github.com/gobabygocarswithjoysticks gobabygocarswithjoysticks@gmail.com

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